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PSL2(Z) and a Four-Tier Obstruction Hierarchy in the Spectral Theory of Polynomial Continued Fractions

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$\mathrm{PSL}_2(\mathbf{Z})$ and a Four-Tier Obstruction Hierarchy in the Spectral Theory of Polynomial Continued Fractions

Working draft

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Abstract

Every arithmetic polynomial continued fraction family, in the sense defined by the current spectral taxonomy, has its hypergeometric Schwarz triangle group commensurable with the full modular group $\mathrm{PSL}_2(\mathbf{Z})$. This is stronger than the bare observation that Takeuchi's classification applies: in the verified arithmetic cases the ambient quotient is always the classical modular curve $X(1)$, rather than an exotic Shimura curve. In particular, every arithmetic case satisfies the algebraic necessary condition $c = 1$, although that condition is not sufficient on its own. We combine this modular theorem with the previously verified Catalan and V_{quad} analyses to obtain a four-tier obstruction hierarchy. Tier 0 consists of genuinely modular families with signature $(\infty, 2, 3)$ but non-CM evaluation points; Tier 1 consists of arithmetic but cuspidal degenerations such as Catalan; Tier 2 consists of non-arithmetic Fuchsian families excluded by Takeuchi's list; and Tier 3 consists of $d = 2$ irregular, confluent-Heun-type families outside the Fuchsian framework altogether. The result is a unified conceptual explanation for the CM no-go phenomena observed across the polynomial continued fraction program.

1 Introduction

The central lesson of the present phase of the PCF program is now clear: the arithmetic families do not merely admit a modular interpretation in some vague sense. They all land on the same

classical modular object. In the verified $d = 1$ hypergeometric sector, every arithmetic family has Schwarz triangle group commensurable with $\mathrm{PSL}_2(\mathbf{Z})$, hence uniformizes the modular curve $X(1)$. This is the most rigid and classical possibility available in Takeuchi's framework [1, 3].

This observation sharpens the earlier CM-point narrative developed across the companion papers in the series [4, 5]. The raw PCF evaluation point z_0 need not itself be a CM value of a modular function; indeed the earlier legs of the computation showed that the naive CM conjecture is false in that direct form. What survives is an ambient modular law: the arithmetic families live on the same modular orbifold, while the observed z_0 is typically a generic, apparently transcendental evaluation point on that orbifold.

The verified dataset now divides into four clean tiers. Tier 0 is the modular sector: arithmetic triangle groups of type $(2, 3, \infty)$, with examples including `log-k=2` and the `pi-m` families. Tier 1 is cuspidal: the Catalan case is arithmetic in origin but collapses to a cusp and acquires a logarithmic Frobenius obstruction. Tier 2 is genuinely non-arithmetic: eight families lie outside Takeuchi's list, so a Shimura-curve CM interpretation is ruled out at the group-theoretic level. Tier 3 is beyond the Fuchsian picture entirely: the $d = 2$ family `V_quad` belongs to a confluent-Heun sector with an irregular singularity at infinity.

The purpose of this draft is therefore twofold. First, it records a concise theorem-level statement of the $\mathrm{PSL}_2(\mathbf{Z})$ phenomenon. Second, it organizes the entire obstruction picture into a four-tier hierarchy that can be used as the conceptual frame for Paper 4.

2 Data and notation

We work with the verified $d = 1$ Gauss-hypergeometric rows produced in the spectral classification workflow. For a witness

$$ {}_2F_1(a, b; c; z), $$

we write the Schwarz period ratio as

$$ \tau(z) = i\,\frac{{}_2F_1(a, b; c; 1 - z)}{{}_2F_1(a, b; c; z)}. $$

following the classical Gauss–Schwarz viewpoint; see, for example, Yoshida's treatment of Schwarz maps and hypergeometric monodromy [2]. In the normalization used throughout the project, the relevant triangle data are extracted from

$$ \lambda = |c - 1|, \qquad \mu = |a - b|, \qquad \nu = |a + b - c|. $$

Thus the corresponding order triple is written as

$$ (\infty, \mu, \nu) $$

whenever $\lambda = 0$, i.e. when one vertex is a cusp.

The computational facts needed in this draft are fully verified:

- there are 15 validated $d = 1$ Gauss rows in the current taxonomy;
- exactly 7 of these are arithmetic in the Takeuchi sense;
- those 7 arithmetic rows all have $(\lambda, \mu, \nu) = (0, 2, 3)$;
- the remaining 8 are non-arithmetic or degenerate;

- Catalan is a cuspidal degeneration with fitted signature $(\infty, 1, 4)$;
- `V_quad` sits in a $d = 2$ irregular sector and is not governed by a Gauss triangle group.

The arithmetic families are displayed in Table 1; note in particular that the third hypergeometric parameter is always $c = 1$ in these rows, so $\lambda = |c - 1| = 0$ identically.

Table 1: Arithmetic PCF families and their verified Schwarz data.

`Family`	*abc*-tuple	(λ, μ, ν)	Uniformization
`log-k=2`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$
`pi-m=0`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$
`pi-m=1`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$
`pi-m=2`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$
`pi-m=3`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$
`pi-m=4`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$
`pi-m=5`	$(-1-i, -1+i, 1)$	$(0, 2, 3)$	$X(1)$

Remark 1 (Why Table 1 repeats the same *abc*-tuple). The identical entries in Table 1 are genuine, not a typographical artifact. All seven Tier 0 arithmetic families share the same local Gauss hypergeometric parameters

$$(a, b, c) = (-1-i, -1+i, 1),$$

so they are governed by the same underlying ${}_2F_1$ differential equation. In the current fitting procedure, the hypergeometric parameters (a, b, c) are determined by the leading coefficients of the recurrence, so families with the same leading-coefficient structure share the same (a, b, c) while differing in their subleading terms and hence in their evaluation points z_0. Thus the families are distinct PCF recurrences with distinct limits and distinct sampled points on the same modular Schwarz curve, even though their local monodromy data agree. In other words, the arithmetic rows are different polynomial continued fraction realizations of a single modular hypergeometric class.

Remark 2 (Same equation, different sampling points). The numerical data illustrate this distinction clearly: for instance, the shared modular witness is sampled at $z_0 \approx 0.2097943444$ for `log-k=2`, at $z_0 \approx 0.3210137317$ for `pi-m=2`, and at $z_0 \approx 0.6627776906$ for `pi-m=5`. Thus the arithmetic families are not seven different Schwarz equations; they are several continued-fraction realizations of one and the same modular Gauss class, observed at different points of the same orbifold.

3 The $\mathrm{PSL}_2(\mathbf{Z})$ theorem

We can now isolate the clean theorem promised by the calculations.

Theorem 1 (Arithmetic PCF families uniformize $X(1)$)**.** *Every arithmetic polynomial continued fraction family in the current verified $d = 1$ taxonomy has hypergeometric Schwarz triangle group commensurable with the full modular group $\mathrm{PSL}_2(\mathbf{Z})$. Equivalently, every arithmetic family uniformizes the classical modular curve $X(1)$.*

Proof sketch. The arithmetic rows are exactly the seven families listed in Table 1. For each of them the fitted hypergeometric data satisfy $c = 1$, hence

$$\lambda = |c - 1| = 0.$$

So one vertex of the Schwarz triangle is necessarily cuspidal. The same verified rows satisfy

$$\mu = |a - b| = 2, \qquad \nu = |a + b - c| = 3,$$

whence the order triple is $(\infty, 2, 3)$. But the orientation-preserving triangle group of type $(2, 3, \infty)$ is the classical modular group generated by

$$T : z \mapsto z + 1, \qquad S : z \mapsto -1/z,$$

that is, $\mathrm{PSL}_2(\mathbf{Z})$. Therefore each arithmetic PCF family is uniformized by the same modular orbifold $\mathrm{PSL}_2(\mathbf{Z})\backslash\mathfrak{H}^* = X(1)$, up to conjugacy or commensurability in the project normalization. □

The theorem should be read as a statement about the *ambient* arithmetic geometry rather than about the numerical value z_0 itself. The earlier CM and Hauptmodul expeditions already showed that the observed z_0 values are not special CM points of the naive type. What Theorem 1 proves is that the only arithmetic PCF families presently seen all lie on the same modular triangle, namely the modular lambda world of $X(1)$.

Remark 3 (Why the condition $c = 1$ matters). The identity $c = 1$ forces $\lambda = 0$ and hence a cusp at one vertex of the Schwarz triangle. The final CSV audit confirms that all seven arithmetic families satisfy $c = 1$ to at least ten-digit precision. However, the converse fails in the current taxonomy: the eight non-arithmetic or degenerate $d = 1$ rows also have $c = 1$. Thus $c = 1$ is a necessary but not sufficient condition for arithmeticity. What distinguishes the arithmetic sector is the sharper pair of constraints $(\mu, \nu) = (2, 3)$, which yields the unique verified modular type $(\infty, 2, 3)$.

4 Worked examples

A useful feature of the new picture is that one can now diagnose the arithmetic type of a family directly from its fitted hypergeometric parameters.

A modular example: `pi-m=2`

For the family `pi-m=2` the verified Gauss parameters are

$$(a, b, c) = (-1 - i, -1 + i, 1).$$

Hence

$$\lambda = |c - 1| = 0, \qquad \mu = |a - b| = 2, \qquad \nu = |a + b - c| = 3.$$

So the signature is exactly $(\infty, 2, 3)$ and Theorem 1 applies. The corresponding period ratio lies in the upper half-plane, numerically near $1.5513679\,i$, which confirms that the family is genuinely modular while still not forcing a special CM value at the observed evaluation point.

A non-arithmetic example: `log-k=3`

For `log-k=3` one again has $c = 1$, so the triangle still has a cusp. However the remaining orders become

$$(\mu, \nu) = (\sqrt{3}, 4),$$

not $(2, 3)$. This single computation already explains why the family is outside the modular Tier 0 regime: it has the right qualitative shape, but the wrong arithmetic angles.

A cuspidal example: Catalan

The Catalan family illustrates the intermediate behavior between these two extremes. Its fitted signature $(\infty, 1, 4)$ shows that the triangle has collapsed beyond the genuine modular regime, and the earlier Frobenius analysis detects the same failure analytically through the persistent $3/(2n)$ residual. This is the prototype for Tier 1.

These three examples already show the logic of the whole classification: the issue is not merely whether there is a cusp, but whether the remaining angular data place the family on the classical modular orbifold, on a degenerate cusp, or outside arithmeticity altogether.

5 A four-tier obstruction hierarchy

The verified classification is best organized as a hierarchy of four distinct obstruction mechanisms. The word "obstruction" is used in a broad sense: in Tier 0 the obstruction is not to modularity itself, but to the stronger hope that the observed PCF evaluation point should automatically be a CM point.

Tier	Geometry	Example	Proved obstruction	Conjectural limit
0	modular $(2, 3, \infty)$	`log-k=2`, `pi-m=0`	generic z_0 is not a CM evaluation point	CM occurs only at sparse distinguished Schwarz points
1	arithmetic but cuspidal	Catalan	Frobenius-log residual and cusp degeneration	all such cases are controlled by cusp data
2	non-arithmetic Fuchsian	`log-k=3`, `log-k=5`	Takeuchi non-arithmeticity	no Shimura reinterpretation exists within Gauss ${}_2F_1$
3	irregular Heun-type	`V_quad`	irregular singularity at infinity	higher-degree PCFs require a non-Fuchsian uniformization

Tier 0: modular families commensurable with $\mathrm{PSL}_2(\mathbf{Z})$

Definition. A family is in Tier 0 if its fitted Schwarz group is arithmetic and has verified signature $(\infty, 2, 3)$. By Theorem 1, every arithmetic family presently in the taxonomy lies in this tier.

Example. The families `log-k=2` and `pi-m=0` are representative. Their hypergeometric witness is of Gauss type with $c = 1$, and the extracted triangle data are exactly $(0, 2, 3)$.

Proved obstruction. The obstruction here is to the original, overly strong CM-point conjecture. The ambient group is modular, but the actual observed value z_0 is a generic point of the Schwarz triangle rather than a special CM coordinate. Thus modularity survives while naive CM evaluation fails.

Conjecture 1 (Tier 0 generic transcendence)**.** *For a Tier 0 family, the PCF evaluation point z_0 is generically transcendental and non-CM, while genuinely algebraic or CM values arise only at isolated modularly distinguished arguments.*

Tier 1: cuspidal arithmetic degeneration

Definition. A family lies in Tier 1 if the underlying geometry is still arithmetic in origin but the hypergeometric or Frobenius data degenerate at a cusp, so the expected holomorphic Schwarz uniformization collapses and logarithmic terms appear.

Example. The Catalan family is the model case. Its fitted Gauss data give the signature $(\infty, 1, 4)$ in the project normalization, and earlier work showed a non-removable logarithmic residual of size $3/(2n)$.

Proved obstruction. There are now two independent proofs. The first, from Paper C, is analytic and Frobenius-theoretic: after normalization one finds a residual term incompatible with the conjectured clean Gauss law. The second, provided by the present paper, is geometric: the Schwarz triangle degenerates cuspidally, so the family never lands in the genuine modular Tier 0 regime.

Conjecture 2 (Tier 1 cusp rigidity)**.** *Every arithmetic-but-degenerate PCF family is governed by a finite list of cusp signatures, and the analytic Frobenius obstruction is determined by that cuspidal degeneration data.*

Tier 2: non-arithmetic Fuchsian obstruction

Definition. A family lies in Tier 2 if it still admits a Gauss hypergeometric witness and hence a three-singularity Fuchsian description, but its triangle signature is not contained in Takeuchi's arithmetic list.

Examples. The current verified examples are the eight non-arithmetic or degenerate rows

`log-k=3`, `log-k=4`, `log-k=5`, `log-k=6`, `log-k=7`, `log-k=8`, `log-k=10`, and `probe-d1-boundary-nn1-3n2`.

Several of these have irrational triangle orders such as $\sqrt{3}$ or $\sqrt{5}$, which already signal non-arithmeticity.

Proved obstruction. In this tier the obstruction is structural and final within the Gauss framework: no family-specific CM or Shimura interpretation of the desired type can exist because the underlying Fuchsian group is not arithmetic.

Lemma 1 (Takeuchi obstruction for non-arithmetic PCF families)**.** *Let $K(a_n, b_n)$ be a $d = 1$ polynomial continued fraction admitting a Gauss hypergeometric witness ${}_2F_1(a, b; c; z)$, and let $\tau(z)$ be its Schwarz period ratio. If the associated triangle-order triple (∞, q, r) with $q = |a - b|$ and $r = |a + b - c|$ does not occur on Takeuchi's arithmetic list, then the Schwarz group is non-arithmetic. Consequently the family has no Shimura-curve CM interpretation of the conjectured type.*

Proof. Takeuchi's theorem classifies arithmetic triangle groups. If the verified order triple of the PCF family is absent from that list, then the associated triangle group cannot be arithmetic. The conjectured CM mechanism requires an arithmetic modular or Shimura uniformization. Hence the family is obstructed at the group-theoretic level. □

Conjecture 3 (Tier 2 rigidity)**.** *No Tier 2 family can be rescued by a rational change of variable, M"obius normalization, or classical Hauptmodul substitution while remaining inside the same Gauss ${}_2F_1$ framework.*

Tier 3: the Heun or irregular sector

Definition. A family lies in Tier 3 if its governing differential equation is no longer a three-regular-singularity Gauss equation, but instead passes to a higher or confluent equation with an irregular singularity. This places the family outside the classical Fuchsian triangle-group world.

Example. The flagship example is `V_quad`. Earlier spectral analysis showed that this family belongs to a clean $d = 2$ tier rather than the $d = 1$ hypergeometric sector; this $d = 2$ verification is consistent with the companion obstruction analysis in Paper C [5].

Proposition 1 (Heun obstruction for `V_quad`)**.** *The family `V_quad` does not admit a Gauss-Schwarz triangle interpretation of the type governing the $d = 1$ arithmetic PCF families. Instead it belongs to an irregular, confluent-Heun-type sector with an obstruction coming from the singularity at infinity.*

Proof sketch. The verified recurrence analysis places `V_quad` in a $d = 2$ clean tier. The associated differential equation is therefore not of the three-regular-singularity Gauss type. In the project interpretation this means the relevant monodromy is no longer described by a Fuchsian triangle group; the singularity at infinity becomes irregular. Hence neither Takeuchi's arithmetic list nor the Tier 0 modular criterion can apply. □

Conjecture 4 (Tier 3 uniformization principle)**.** *Tier 3 families should admit a different arithmetic or geometric description, if any exists, through a Heun, confluent-Heun, or isomonodromic framework rather than through classical Schwarz triangles.*

6 Catalan and the connection to Paper C

The Catalan family deserves separate emphasis because it is the first place where the four-tier hierarchy can be seen from two independent directions. The earlier submission usually referred to internally as Paper C (submission 268704746) isolated the Catalan obstruction analytically through the persistent Frobenius residual $3/(2n)$ [5]. That argument showed that no clean Gauss identity survives after normalization.

Remark 4 (Independent second proof for Catalan)*.* The Catalan obstruction was first proved in Paper C via the $3/(2n)$ Frobenius residual. The present work provides an independent proof via the cuspidal degeneration of the Schwarz triangle group. The agreement of these two arguments supplies strong structural evidence for the four-tier hierarchy.

This is conceptually important. A single computational anomaly could always be blamed on a bad normalization or an accidental numerical fit. Here, by contrast, the same family is obstructed both analytically and geometrically. The Frobenius defect and the cuspidal triangle signature describe the same phenomenon from different sides.

7 Consequences for the CM-point program

The hierarchy above organizes the entire CM-point narrative into a precise logical flow.

1. If a family is in Tier 0, then it is genuinely modular, but one should not expect the observed PCF evaluation point to be CM without additional structure.

2. If a family is in Tier 1, then arithmeticity persists only in degenerate, cuspidal form; one expects logarithmic or Frobenius residues rather than clean modular identities.

3. If a family is in Tier 2, then the CM hope fails for a definitive group-theoretic reason: the Fuchsian monodromy is non-arithmetic.

4. If a family is in Tier 3, then the very language of Fuchsian triangles is no longer the correct one, so a different theory must replace the modular-Shimura picture.

In this sense the outcome of the expedition is not a single no-go theorem but a taxonomy of failure modes. Such a taxonomy is mathematically stronger than a collection of isolated negative computations, because it predicts what kind of obstruction should occur before one even starts a detailed fit.

Conjecture 5 (Master hierarchy conjecture)**.** *Every polynomial continued fraction family in the present spectral program admits a well-defined tier within the framework of Section 5. Moreover, the tier determines the correct ambient geometry: modular, cuspidal, non-arithmetic Fuchsian, or irregular Heun.*

8 Relation to the companion series

The present manuscript is intended to serve as the number-theoretic capstone of the current four-paper program on polynomial continued fractions. Paper A established the computational and taxonomic framework in which the $d = 1$ and $d = 2$ families could be separated and studied systematically [4]. Paper C then isolated the Catalan Frobenius obstruction and clarified the degenerating and higher-order behaviors that now appear as Tiers 1 and 3 [5].

The new contribution of Paper 4 is to extract the arithmetic theorem hidden inside that experimental infrastructure. Instead of presenting another isolated discovery, it shows that the verified families satisfy a sharp modular dichotomy: arithmetic examples all return to the single orbifold $X(1)$, while the non-modular examples fail for structurally different reasons. In this way the four-tier hierarchy packages the earlier computations into a form closer to the language of number theory and automorphic geometry.

9 Outlook

From the standpoint of number theory, the striking discovery is that all currently arithmetic PCF families collapse onto the same simplest modular curve $X(1)$. This makes the phenomenon both narrower and more elegant than initially expected. The families do not scatter across a zoo of Shimura curves; they all return to the classical modular group.

From the standpoint of continued fractions, the four-tier hierarchy provides a practical guide for future searches. One should expect modular identities only in Tier 0, degeneration phenomena in Tier 1, arithmetic impossibility in Tier 2, and new differential equations in Tier 3. In particular, the next conceptual frontier is not another blind CM search, but a controlled exploration of Heun-type and higher-rank sectors.

Summary. The verified evidence now supports a concise thesis: the arithmetic $d = 1$ polynomial continued fractions are all modular of $\mathrm{PSL}_2(\mathbf{Z})$ type, while every failure of the naive CM-point conjecture can be placed into one of three further obstruction tiers. This is exactly the kind of structural packaging needed for a self-contained Paper 4.

Declaration on the use of AI tools

The computational experiments, PSLQ verification runs, spectral taxonomy, and iterative proof development underlying this paper were carried out using a multi-agent research workflow combining three AI-assisted environments: GitHub Copilot (within Visual Studio Code) for autonomous code generation, iterative script refinement, and workspace-level memory across sessions; and Anthropic Claude for high-level strategic guidance, mathematical peer review, prompt engineering, and synthesis of results across the research program. All mathematical claims, proofs, and numerical verifications were independently confirmed by the author. The Python scripts and CSV data supporting the results are publicly available at https://github.com/papanokechi/pcf-spectral-classes.

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